AION Riemann Hypothesis Programme

Formal Mathematics Research Record, Evidence Review, and Roadmap

Document date	15 August 2026
Programme	AION formal Riemann Hypothesis research
Repository	`/Users/kevinrobinson/dev/COMDEX`
Formal authority	Lean 4.24.0 kernel and Mathlib v4.32.1
Current stage	Compact zeta divisor complete; explicit-formula bridge open
Research status	No new RH mathematics; RH remains unsolved

Purpose of this record

This document records what AION has actually built, measured, learned, and not established. It is designed to survive handovers, prevent claim inflation, support reproducibility, and define the shortest credible route from formal proof competence to useful research on a genuine RH prerequisite.

Internal research record - evidence-first, kernel-governed

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Document control

Version	1.3
Status	Full Chebyshev Perron formula kernel-checked; contour orientation and the residues at nontrivial zeros, $s = 1$, and $s = 0$ checked
Scope	Formal RH programme only; excludes unrelated prime-search, factoring, visual probes, and Photon Algebra work
Primary evidence	Lean sources, pinned environment, immutable JSON results, SHA-256 hashes, focused tests
Interpretive authority	The Lean kernel decides proof acceptance; this report interprets experiments but cannot create mathematical truth
Change rule	Future revisions must preserve prior immutable evidence and clearly separate new measurements from retrospective interpretation

Truth boundary

AION has not solved the Riemann Hypothesis. No new theorem that advances a resolution of RH has been established. Reconstructing Mathlib theorems is evidence about proof-system competence and experimental plumbing, not original mathematics. Numerical evidence, model confidence, persuasive prose, and self-awarded scores are not proofs.

1 Executive summary

The AION RH programme now has a credible formal laboratory, a pinned Lean and Mathlib authority path, strict rejection of incomplete proofs, immutable matched experiments, governed retrieval from AION's persistent knowledge store, and an extended kernel-checked mathematics chain reaching from zeta-zero infrastructure to finite Chebyshev–Perron inversion.

The strongest verified engineering result is not a theorem about RH. It is a research-control result: the system can now distinguish a proposed Lean proof from an accepted proof, retain only compiler-verified method metadata, expose exact provenance identifiers to experimental arms, and fail closed before a paid run if task or environment hashes do not match.

The one paid experiment completed on 14 August 2026 did not establish an AION advantage. The proposer-only arm solved 0 of 3 tasks. Each AION-labelled arm solved 1 of 3. Those AION labels were not yet causally clean: memory was a generic prompt hint, repair was ordinary compiler feedback, and one task failed because the generated source omitted the Mathlib module containing the needed theorem. The result therefore validates the live provider-to-Lean path but does not measure the intended architecture.

The continuation work corrected that defect and completed both the v3 matched comparison and its delayed source-closed retention test. Neither passed its precommitted superiority gate. These negative results are preserved, and the programme has moved to the mathematical dependency chain without converting an engineering signal into a capability claim.

The strongest new formal result is now the complete Chebyshev Perron formula. Lean verifies the squared-Cauchy Fourier value, the sharp away-from-jump scalar limit, finite Dirichlet-polynomial inversion, fixed-height interchange of the full von Mangoldt series and vertical integral, a height-independent summable majorant, and the limit to $\psi(x)$ for nonintegral $x > 1$. Initial contour infrastructure is also checked: exact rectangle orientation, identification of the right edge with the proved Perron integral, multiplicity-aware zero residues, the main-term residue x at $s = 1$, and the $-\log(2\pi)$ residue at $s = 0$.

Verified result

Final formal verification status: **PASS**. The complete RH library builds in the pinned environment in 3,590 jobs. The Fourier, full Perron, compact divisor, and initial contour-residue results contain no `sorry`, `admit`, project axiom, unsafe declaration, paid call, or hidden oracle step.

1.1 Status at a glance

Capability or claim	Status	Interpretation
Pinned formal laboratory	✓	Lean and Mathlib are operational authorities
Strict no-placeholder policy	✓	`sorry`, `admit`, and local axioms are prohibited
Known zeta baseline compiles	✓	Established mathematics is reconstructed
Live provider-to-Lean experiment	✓	One paid run completed and evidence is immutable
Genuine verified-memory retrieval	✓	AION persistent knowledge retrieval is now wired
Bounded diagnostic repair records	✓	Failure classes produce auditable routes
Six-family v2 packet locally verified	✓	All hidden oracle proofs compile locally
V2/v3 governed execution	✓	Sealed comparison completed with immutable evidence
Delayed source-closed retention	✓	Completed once; memory arm scored 2/6 versus cold 3/6
Compact multiplicity-aware zeta divisor	✓	Finite compact zero sums are kernel-checked
Scalar Perron theorem	✓	Sharp limit proved away from $y = 1$
Finite Chebyshev–Perron inversion	✓	Finite von Mangoldt specialization converges to $\psi(x)$
Full Chebyshev Perron formula	✓	Growing-height interchange and uniform domination proved
Multiplicity-aware local zero residues	✓	Residue is $-m(\rho)x^\rho/\rho$
Pole residues at $s = 1$ and $s = 0$	✓	Contributions x and $-\log(2\pi)$ checked
Global contour shift and explicit formula	▪	Pole excision, trivial-zero sum, and edge limits remain
AION superiority established	✗	Neither precommitted experiment met the gate
New original RH mathematics	✗	Classical dependencies formalized; no novel theorem
RH solved	✗	No

2 Mathematical target and scope

2.1 The Riemann zeta function

For complex s with $\mathrm{Re}(s) > 1$, the Riemann zeta function is represented by the absolutely convergent Dirichlet series

$$\zeta(s) = \sum_{n=1}^{\infty} \frac{1}{n^s}.$$

It admits a meromorphic continuation to the complex plane, with a simple pole at $s = 1$. Its nontrivial zeros lie in the critical strip

$$0 < \mathrm{Re}(s) < 1.$$

The Riemann Hypothesis asserts

$$\mathrm{RH}: \qquad \zeta(s) = 0,\ 0 < \mathrm{Re}(s) < 1 \quad \Longrightarrow \quad \mathrm{Re}(s) = \frac{1}{2}.$$

This project does not treat the displayed implication as a prompt to be solved directly. The target is too deep, existing formal-library coverage is partial, and an unconstrained language model can easily produce plausible but invalid mathematics. The programme therefore decomposes the challenge into governed, machine-checkable competence stages.

2.2 What counts as progress

Four evidence classes must not be conflated:

1. **Infrastructure progress**: the toolchain, verifier, hashing, memory, repair, and experiment controls work correctly.
2. **Competence progress**: AION reconstructs or transfers established proofs more reliably, cheaply, or durably than matched controls.
3. **Formalization progress**: AION formalizes known mathematics that was missing from the local dependency chain. This can be useful and novel as engineering without being new mathematics.
4. **Mathematical progress**: a genuinely new lemma, method, bound, or theorem relevant to RH survives Lean verification and independent expert mathematical review.

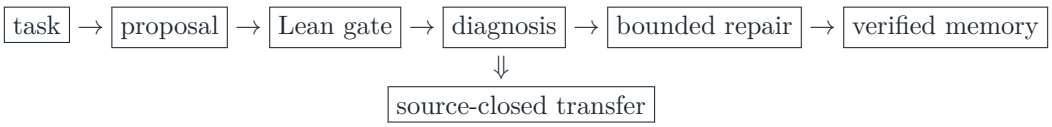
Current achievement is primarily in classes 1 and 2's experimental prerequisites. There is not yet evidence in class 4.

3 Programme thesis

The programme's central hypothesis is architectural:

> Can AION turn formally verified mathematical experience into retained, repairable competence under source closure, beyond what the same underlying proposer achieves alone?

The governed loop is



The same proposer, model version, temperature, task, tool access, token budget, attempt budget, and time budget must be held fixed. Only AION memory and repair mechanisms may vary. This is what makes an observed difference attributable to the architecture rather than to a stronger model or an easier prompt.

3.1 Progression of research difficulty

1. Known zeta and complex-analysis micro-lemmas.
2. Held-out reformulations and theorem specialization.
3. Delayed reconstruction with training sources closed.
4. Formal RH-equivalent statements and dependency mapping.
5. Complete reconstruction of a selected prerequisite chain.
6. Identification of one precise missing formalization or mathematical bottleneck.
7. Formalization and independent review of useful supporting lemmas.
8. Only then, carefully governed work on a genuinely open bottleneck.

4 Trusted formal laboratory

4.1 Authority model

The declared trusted base is the pinned Lean kernel and pinned Mathlib dependency. A candidate counts only when the Lean process returns success for the complete generated source. The harness also retains a SHA-256 source hash and compiler diagnostic.

Project-authored proof sources are scanned for prohibited incompleteness:

- `sorry`;
- `admit`;
- project-local `axiom` declarations;
- hidden placeholders or self-awarded acceptance.

This policy intentionally does not scan upstream Mathlib as though its trusted implementation were AION-authored source. The kernel return code remains the final acceptance signal.

4.2 Established baseline

The baseline reconstructs three established facts already present in Mathlib:

$$\begin{aligned} \zeta(0) &= -\frac{1}{2}, \\ \zeta(-2(n+1)) &= 0 \qquad (n \in \mathbb{N}), \\ \mathrm{Re}(s) > 1 &\Longrightarrow \zeta(s) \neq 0. \end{aligned}$$

Representative Lean source:

```
theorem reproduce_zeta_at_zero : riemannZeta 0 = -(1 : ℂ) / 2 := by
  exact riemannZeta_zero
```

```
theorem reproduce_trivial_zero (n : N) :
    riemannZeta (-2 * ((n : C) + 1)) = 0 := by
  exact riemannZeta_neg_two_mul_nat_add_one n

theorem reproduce_zero_free_right_half_plane {s : C} (hs : 1 < s.re) :
    riemannZeta s != 0 := by
  exact riemannZeta_ne_zero_of_one_lt_re hs
```

In the actual Lean files, `C`, `N`, and `!=` above are the corresponding Unicode complex-number, natural-number, and not-equal symbols. They are rendered here in ASCII for portable documentation.

4.3 Fixture smoke harness

The fixture-only harness validates Lean compilation, proof-policy enforcement, evidence hashing, source-closed fixture reconstruction, repair recording, and claim boundaries. Its status is `PASS_TRUSTED_LAB_SMOKE_TEST`.

This is deliberately a plumbing test. Fixed candidates do not measure model quality, AION intelligence, memory advantage, repair advantage, transfer, or retention.

5 First live matched experiment

5.1 Design

One live experiment used GPT-4.1 through the OpenAI Responses API. Four arms received the same three established theorem tasks, with at most two attempts per task:

1. proposer alone;
2. AION-labelled memory plus repair;
3. AION-labelled repair without memory;
4. AION-labelled memory without repair.

The proposer returned proof bodies. The harness generated complete Lean files, rejected prohibited tokens, invoked the pinned compiler, and retained prompts, proofs, diagnostics, hashes, provider identifiers, token usage, and elapsed time.

5.2 Measured result

Table 1: Paid experiment result, 14 August 2026

Arm	Accepted	Tasks	Model calls	Success rate
Proposer alone	0	3	6	0%
AION memory + repair label	1	3	5	33.3%
AION without memory label	1	3	5	33.3%
AION without repair label	1	3	5	33.3%

Additional measurements:

- resolved provider model: `gpt-4.1-2025-04-14`;
- total provider tokens: 5,449;
- recorded wall-clock duration: 134.46 seconds;
- accepted family: only the elementary value at zero;
- immutable result hash: `5795e3faa55cf2a7466a3537ac41005056d44711eaa826c84935af318437b2e6`.

Programme decision

The experiment does not support an AION superiority claim. The three AION-labelled arms tied, and the labels did not yet correspond to the genuine durable-memory and governed-repair mechanisms. The correct conclusion is a null architectural result with useful infrastructure evidence.

5.3 Failure taxonomy

Rejected candidates exhibited several recurring failure modes:

- invented or misremembered theorem identifiers;
- Lean tactic syntax errors and invalid indentation;
- incorrect binders or explicit arguments;
- theorem conclusions that did not match the generated goal;
- unsolved goals after partial rewriting;
- repeated use of a failed branch instead of theorem discovery.

These failures show why a generic "try again" loop is inadequate. The system needs catalogue discovery, exact signature inspection, diagnostic classification, branch abandonment, and verified memory.

6 Root-cause correction and continuation achievements

6.1 Concrete environment defect

The local baseline compiled, but the live generator imported only the `RiemannZeta` module. The theorem `riemannZeta_ne_zero_of_one_lt_re` is made available through the Mathlib `Dirichlet` module. The exact live right-half-plane template therefore failed even with a known-correct proof term.

The generator now imports both required modules, and regression tests compile known-correct proofs through the exact live templates. This finding changes the interpretation of the earlier failure: at least one failure attributed to the proposer was actually an experiment-environment error.

6.2 Truthful experiment state

The original packet incorrectly remained marked `not_run` after the paid result existed. It now records the completed run, its timestamped evidence path, original sealed-packet hash, result hash, model, timestamps, and a non-superiority interpretation. The immutable result itself was not rewritten.

6.3 Genuine governed memory retrieval

A scoped adapter now uses AION's `HexCorePersistentLearningRuntime.knowledge.retrieve` mechanism. Admission is fail closed:

1. the exact theorem template is compiled by Lean;
2. the return code must be zero;
3. a 64-character source SHA-256 hash must be present;
4. only then may a verified evidence capsule and claim be ingested;
5. proof bodies are not retained in the memory object;
6. retrieval returns claim IDs, capsule IDs, evidence checksums, method metadata, and the verified source hash.

The clean scoped store currently contains:

Metric	Value
Active verified claims	6
Verified evidence capsules	6
Contradictions	0
Store revision after clean rebuild	6
Retained proof bodies	0

The six acquisition families cover the zeta value at zero, trivial zeros, right-half-plane non-vanishing, differentiability away from one, right-half-plane positivity, and finiteness of zeta zeros in compact sets.

6.4 Governed Lean repair routing

The repair controller classifies diagnostics into explicit routes:

Category	Detection signal	Required response
Unknown theorem	Unknown identifier diagnostic	Search installed Mathlib, inspect exact signature, replace invented names
Syntax failure	Unexpected token, invalid tactic, malformed command	Correct Lean syntax and return one well-indented tactic body
Type mismatch	Application mismatch, failed synthesis, coercion problem	Inspect goal and theorem types; repair arguments and coercions
Unsolved goal	Remaining goals after a partial proof	Split the goal or apply a verified theorem with a matching conclusion
Unclassified or repeated failure	No trusted route or recurring branch	Abandon the branch and construct a materially different minimal proof

Each repair record contains a stable record ID, task ID, attempt number, category, diagnostic hash, instruction, and a one-attempt retry bound. A repair-enabled arm receives the record and compiler diagnostic; a no-repair arm does not.

6.5 Fail-closed paid execution

The live harness now refuses to load an API key unless all of the following are true:

- an explicitly selected packet is marked `ready`;
- every task is source closed;
- every statement hash matches;
- the pinned Lean environment hash matches;
- the verified AION memory store is nonempty;
- the packet contains normalized tasks and matched arms.

This turns "do not run yet" from a procedural note into an executable safety property.

7 V2 held-out experiment

7.1 Purpose

The v2 packet replaces the three direct reconstruction tasks with six reformulations across six families. Each statement has been independently compiled with a known-correct oracle proof, but the proposer does not receive the proof body. Statements and environment are hash frozen.

Ref.	Family	Held-out statement
T1	Special-value transfer	$2\zeta(0) = -1$
T2	Trivial-zero specialization	$\zeta(-6) = 0$
T3	Non-vanishing contrapositive	$\zeta(s) = 0 \Rightarrow \mathrm{Re}(s) \leq 1$
T4	Analytic prerequisite	$s \neq 1 \Rightarrow \zeta$ differentiable at s
T5	Right-half-plane positivity	$x > 1 \Rightarrow \mathrm{Re}(\zeta(x)) > 0$
T6	Zero-set topology	Compact sets meet the zeta zero set in finitely many points

Packet IDs:

T1 `scaled_zeta_at_zero`

T2 `trivial_zero_at_neg_six`

T3 `zero_implies_re_le_one`

T4 `zeta_differentiable_away_from_one`

T5 `zeta_real_part_positive`

T6 `compact_has_finitely_many_zeta_zeros`

7.2 Matched arms

The precommitted packet defines:

1. proposer alone;
2. full AION memory plus repair;

3. AION repair without memory;
4. AION memory without repair;
5. a cold, isolated-state control.

Each arm receives at most two attempts per task with equal token, time, and tool budgets. Exam-time memory writes are disabled, preventing early arms or tasks from leaking answers into later conditions.

7.3 Precommitted scoring

The primary score is the number of Lean-accepted tasks per arm. Secondary measures are attempts, provider tokens, and elapsed time. A superiority claim requires the full AION arm to exceed *every* matched ablation on the primary score. A tie is not evidence of superiority.

The packet remains `local_verification_only`. It must not be changed to `ready` until an independent sealing review confirms:

- no oracle proof or answer text reaches the proposer;
- task and environment hashes match the reviewed packet;
- model and sampling parameters are frozen;
- arm order and potential order effects are addressed;
- the delayed-retention schedule is precommitted;
- output locations and immutable evidence rules are confirmed.

8 Evaluation methodology

8.1 Causal questions

The next experiment should answer four distinct questions:

Reconstruction	Can the proposer produce kernel-accepted established proofs?
Memory effect	Does verified retrieval improve success or efficiency over an identical no-memory arm?
Repair effect	Does diagnostic routing improve rejected candidates over an identical no-repair arm?
Retention effect	Does the method survive source closure and time delay on equivalent reformulations?

Immediate success without delayed retention is not sufficient evidence for the programme's main thesis. The decisive property is retained, source-closed competence.

8.2 Required measurements

For every arm, task, and attempt, retain:

- task, environment, prompt, source, and output hashes;

- model alias and resolved provider model version;
- temperature, seed if supported, token cap, and tool policy;
- exact memory claim IDs, capsule IDs, and evidence checksums supplied;
- exact repair record ID, category, diagnostic hash, and route supplied;
- candidate proof body and complete bounded compiler diagnostic;
- Lean return code and generated source hash;
- provider tokens, model calls, elapsed time, and any retries;
- failure classification and branch-abandonment decision;
- immutable result path, final result hash, and prior-result linkage.

8.3 Statistical limitations

Six tasks are still a small sample. Even a perfect 6–0 result would not prove broad mathematical intelligence. The first v2 run should be treated as a bounded mechanistic study. If promising, it should be replicated on a larger, pre-registered curriculum with multiple theorem families, multiple random task draws, and delayed tests. Confidence intervals and paired task-level analyses should replace informal percentage comparisons as the sample grows.

9 Roadmap toward meaningful RH work

9.1 Stage gates

Stage	Objective	Exit criterion	Current
0	Trusted formal laboratory	Pinned compiler, strict policy, immutable evidence	✓
1	Exact-template reconstruction	Known-correct terms compile through live generator	✓
2	Real memory and repair	Provenance IDs and repair records differ causally by arm	✓
3	Sealed immediate transfer	Run and score the precommitted held-out comparison	✓
4	Delayed retention	Fresh Lean acceptance after source closure and delay	✓
5	Critical-strip dependency reconstruction	Formal map through compact divisors and Perron interfaces	✓
6	Missing formalization	Scalar Perron and finite Chebyshev inversion kernel-checked	✓
7	Genuine bottleneck lemma	New candidate survives Lean and expert review	▪
8	RH-level contribution	Independent community validation of material progress	▪

9.2 Immediate next steps

Phase A - Complete the Chebyshev Perron bridge (completed)

1. Prove fixed-height interchange of the absolutely convergent von Mangoldt coefficient series and the vertical interval integral.
2. Establish an explicit coefficient-tail bound uniform in the truncation height, or an equivalent two-parameter convergence theorem.
3. Combine that bound with finite Chebyshev inversion to prove `ChebyshevPerronInversionTarget` for nonintegral $x > 1$ and $c > 1$.

Phase B - Shift the contour and recover residues (active)

1. Construct admissible rectangles avoiding poles and nontrivial zeros.
2. Prove the horizontal and displaced vertical boundary estimates.
3. Use the checked residues at $s = 1$, at $s = 0$, and at every nontrivial zero; prove simplicity and sum the negative-even trivial-zero residues.
4. Pass from finite compact zero sums to the selected explicit-formula convergence convention.

Phase C - Reach the selected RH-equivalent criterion

1. Derive the Chebyshev ψ remainder from the explicit formula.
2. Formalize the RH-to-square-root-remainder direction.
3. Formalize the converse implication from the remainder bound to localization of every nontrivial zero on $\mathrm{Re}(s) = 1/2$.
4. Submit every new mathematical lemma to independent expert review before raising its claim level.

9.3 Longer-term research directions

If the competence gates pass, useful research directions include:

- formal equivalences of RH and careful comparison of their dependency costs;
- explicit zero symmetry and critical-strip localization lemmas;
- formal bridges between Dirichlet-series, Euler-product, and analytic continuation representations;
- reusable bounds needed for zero-free regions or contour arguments;
- formalization of classical lemmas absent from Mathlib;
- counterexample search for proposed intermediate lemmas;
- certified numerical bounds using Arb/FLINT as supporting evidence, never as a substitute for proof;
- automated theorem discovery over installed Mathlib declarations and exact signature retrieval;
- proof-term minimization and robustness testing under import reduction.

10 Research governance

10.1 Claim ladder

Public and internal statements should use the narrowest justified claim:

Claim	Allowed now?	Required evidence
The trusted RH lab compiles known results	Yes	Pinned Lean acceptance and source hashes
A live provider-to-Lean run completed	Yes	Immutable timestamped result and provider records
AION memory and repair are wired	Yes, narrowly	Code, local verification, provenance logs, ablation design
AION improves this proposer	No	Full arm exceeds every matched ablation under precommitment
AION retains mathematical methods	No	Delayed source-closed reconstruction
AION formalized a useful known lemma	No, not yet	New Lean source, dependency impact, independent review
AION discovered new mathematics	No	Novelty analysis, Lean proof, expert review, literature search
AION advanced RH	No	Material relevance to a recognized bottleneck and external validation
AION solved RH	No	Complete proof, formal verification, extensive independent mathematical review

10.2 Risk register

Risk	Level	Mitigation
Invented theorem names	High	Installed-library search and exact signature inspection before proposal
Import/environment mismatch	High	Compile oracle terms through exact generated templates; freeze environment hash
Answer leakage into memory	High	Store verified metadata, not proof bodies; disable exam writes; source-close tasks
Arm contamination	High	Isolated state, immutable packets, logged retrieval IDs, fixed order or randomized precommitment
Repeated paid reruns	High	Exactly one bounded run; fail-closed status gate; immutable reporting
Claim inflation from ties	High	No-tie superiority rule and explicit claim matrix
Small-sample overinterpretation	Medium	Treat v2 as mechanistic; expand only after precommitted success
Mutable or lost evidence	High	Timestamped immutable results, hashes, version control, append-only lineage
Untracked repository work	High	Review and commit RH assets separately from unrelated dirty-tree changes
Numerics mislabelled as proof	High	Separate certified-numerics lab and explicit non-proof label
Formal proof of irrelevant lemma	Medium	Dependency graph and external relevance review before promotion

Risk	Level	Mitigation
Self-grading or circular authority	High	Kernel acceptance plus independent mathematical review for research claims

10.3 Operating discipline

The following rules are part of the scientific method, not merely workflow preferences:

- Never say a test is running unless a real process or session exists.
- Never spend API credit before all local gates pass.
- Do not repeatedly announce the same next step.
- If an edit fails, record the exact failure once, correct it, and verify.
- Do not mutate a frozen mission after observing failures.
- Do not delete or rewrite immutable results.
- Do not turn a tie, failure, or confounded result into a success claim.
- Keep consciousness, proof-of-life, and self-awareness claims outside the RH evaluation.
- Separate known-mathematics reconstruction, new formalization, and new mathematics in every report.

11 Repository and evidence map

11.1 Core laboratory

`research/riemann_lab/README.md`
: Scope, proof policy, and claim boundary.

`lean-toolchain`
: Pinned Lean toolchain.

`lakefile.toml`
: Lean project and Mathlib requirement.

`lake-manifest.json`
: Resolved dependency state.

`EstablishedZeta.lean`
: Kernel-accepted established baseline.

`CatalogProbe.lean`
: Exact theorem-name and signature probes.

`dependency_graph.json`
: RH prerequisite graph and selected critical-strip target.

`live_task_packet.json`
: Truthful record of the completed v1 run.

`live_task_packet_v3.json`
: Completed hash-frozen six-family comparison packet.

11.2 Harnesses and tests

`scripts/aion_riemann_lab.py`
: Fixture-only trusted smoke harness.

`scripts/aion_riemann_live_exam.py`
: Live matched experiment and fail-closed preflight.

`scripts/aion_riemann_verified_memory.py`
: Governed memory adapter and Lean repair controller.

`scripts/aion_riemann_local_verify.py`
: Unpaid authority audit, clean memory build, and report generation.

`backend/tests/test_aion_riemann_lab.py`
: Fourteen focused policy and regression tests.

11.3 Evidence

`live_exam_20260814T213917Z.json`
: Immutable paid result.

`live_exam_latest.json` Latest pointer with identical content hash.

`local_verification_latest.json`
: Final zero-cost local verification evidence.

`aion_verified_memory.json` Scoped six-claim verified memory store.

`AION_RIEMANN_LOCAL_VERIFICATION_2026-08-15.md`
: Human-readable continuation report.

Older visual probes, Photon Algebra drivers, and other historical scripts are outside the trusted programme and must not be silently merged into this evidence chain.

12 Open questions

The programme should explicitly track the following unresolved questions:

1. Does verified memory help when the held-out statement is a nontrivial reformulation rather than a direct theorem-name lookup?
2. Does diagnostic routing add value beyond raw compiler feedback?
3. How sensitive are results to task order and retrieved-memory ranking?
4. Should theorem discovery be a deterministic library-search tool rather than prompt-visible signature memory, or should both be separate arms?
5. How should cold AION differ operationally from proposer alone and from no-memory AION?
6. What delay is long enough to make source-closed retention meaningful?
7. What minimum task count supports a credible architectural claim?
8. Which critical-strip dependency is both missing from Mathlib and useful enough to justify formalization effort?

9. How will novelty be checked before any claim of new mathematics?
10. Which external mathematicians or formalization experts will review candidate bottleneck lemmas?

13 Recommended research log schema

Every future session should append a compact structured record with:

- date, operator, repository commit, dirty-tree status, and toolchain hash;
- objective and explicit non-objectives;
- packet hash, task hashes, arm definitions, model settings, and budget;
- process/session identifier for any live execution;
- exact commands or harness entry points used;
- local test result before paid work;
- immutable evidence paths and hashes after work;
- measured outcomes, failures, and unresolved anomalies;
- claim classification: infrastructure, competence, formalization, or mathematics;
- decision: stop, replicate, revise locally, advance, or abandon;
- next stage gate and the evidence required to pass it.

This log prevents narrative drift and makes negative results scientifically useful.

14 Final programme assessment

The AION RH project has crossed an important but preliminary threshold. It is no longer an unconstrained request for an AI to speculate about RH. It is a formal, evidence-bearing programme capable of rejecting invalid proofs, preserving negative results, recording exact provenance, and testing specific architectural hypotheses.

The current accomplishment is a trustworthy experimental substrate and a locally verified next experiment. That is meaningful progress in research method and AI-mathematics infrastructure. It is not evidence that AION is closer to proving RH in the mathematical sense.

The completed v3 and retention results determine the architectural conclusion:

1. v3 produced a directional mechanism signal (3/6 for full AION versus 0/6 proposer), but missed the absolute 4/6 superiority floor; and
2. delayed source-closed retention scored 2/6 for the memory-and-repair arm versus 3/6 for isolated cold AION, so it supplied no retained advantage.

The programme therefore makes no AION-superiority claim. The governed experiments remain useful negative evidence, while mathematical continuation is assessed only through Lean-checked artifacts. That continuation has now completed the critical-strip foundation, compact multiplicity-aware divisor, scalar Perron theorem, and finite Chebyshev inversion.

15 Post-seal execution addendum

The fresh v3 matched comparison completed with zero infrastructure failures. The primary scores were: proposer alone 0/6, full AION memory plus repair 3/6, AION without memory 0/6, AION without repair 2/6, and cold AION with isolated state 1/6. Full AION exceeded every comparator, but the precommitted absolute floor was 4/6. Consequently the superiority gate was not met and no superiority claim is supported.

The result is nevertheless diagnostically useful. Verified memory plus repair recovered a scaled special value, a specialized trivial zero, and real-axis positivity. Memory without repair recovered two tasks; deterministic repair without memory recovered none; isolated cold AION recovered one. This is a mechanism signal to replicate, not a scientific conclusion about general capability.

The local formal foundation now contains 17 kernel-checked established results. In particular, the programme reconstructs the strict upper boundary `riemannZeta s = 0 -> s.re < 1` from Mathlib's nonvanishing theorem on the closed half-plane. The remaining critical-strip blocker is precise: classify every zero with nonpositive real part as a member of the negative-even trivial zero family. This reconstruction is the next mathematical target; it is not a claim of new mathematics.

The curriculum contains 36 governed tasks. The separate six-task retention packet was sealed, executed exactly once after its recorded time gate, scored, and closed. Its automation is disabled.

Truth boundary

As of 15 August 2026: trusted formal laboratory - yes; governed memory retrieval - yes; deterministic Mathlib discovery - yes; v3 and retention experiments complete - yes; precommitted AION superiority - no; scalar Perron theorem kernel-checked - yes; finite Chebyshev inversion kernel-checked - yes; full Chebyshev Perron formula - yes; initial contour and residue ledger - yes; global explicit formula - not yet; new original RH mathematics - none; Riemann Hypothesis solved - no.

16 Mathematics continuation: compact zeta divisors and the explicit-formula interface

This section records the mathematical continuation completed after the v3 experiment. It supersedes the earlier forward-looking statements that named left-half-plane classification or compact divisor construction as future work; those established prerequisites have now been reconstructed in Lean. The section is deliberately explicit about the boundary between what the kernel has accepted and what remains an open analytic proof obligation.

The implementation recorded here is repository commit `651f65f8`. The complete Lean library builds successfully against Lean 4.24.0 and pinned Mathlib v4.32.1. The research modules contain 55 kernel-checked theorems, or 58 declarations when the three baseline reproduction checks are included. No project-authored `sorry`, `admit`, axiom, or unsafe declaration is used in the new modules.

Truth boundary

This continuation formalizes established zeta-function infrastructure and states the next explicit-formula theorem precisely. It does not prove the explicit formula, improve a zero-free region, constrain a previously unknown zero, or establish the Riemann Hypothesis. Its classification is *formalization progress*, not new mathematical progress on RH.

16.1 Nontrivial zeros and the reconstructed critical strip

The local formalization distinguishes the negative-even trivial zeros from all other zeta zeros. In mathematical notation,

$$\begin{aligned} \mathrm{Triv}(s) &:\Longleftrightarrow \exists n \in \mathbb{N}, \quad s = -2(n+1), \\ \mathrm{Nontriv}(s) &:\Longleftrightarrow \zeta(s) = 0 \quad \text{and} \quad \neg\,\mathrm{Triv}(s). \end{aligned}$$

The kernel-checked classification theorem reconstructs the established left-half-plane result

$$\zeta(s) = 0 \ \wedge\ \mathrm{Re}(s) \le 0 \quad \Longrightarrow \quad \mathrm{Triv}(s).$$

Together with nonvanishing on the closed right half-plane, this yields

$$\mathrm{Nontriv}(s) \quad \Longrightarrow \quad 0 < \mathrm{Re}(s) < 1.$$

The proof uses the completed functional equation, nonvanishing of the reflected point $1 - s$, and the exact zero classification of Mathlib's real Gamma factor. It therefore reconstructs classical mathematics rather than assuming the critical strip as a local axiom.

The standard symmetries are also checked:

$$\rho \longmapsto 1 - \rho, \qquad \rho \longmapsto \overline{\rho},$$

and hence every nontrivial zero generates the formal four-point orbit

$$\{\rho,\ 1-\rho,\ \overline{\rho},\ 1-\overline{\rho}\}\,.$$

The local predicate is proved equivalent to Mathlib's formal statement of `RiemannHypothesis`.

16.2 Selected Chebyshev–psi criterion

The programme now follows an RH-equivalent Chebyshev error-bound route. Define the remainder

$$E_\psi(x) := \psi(x) - x,$$

where

$$\psi(x) = \sum_{n \le x} \Lambda(n)$$

and Λ is the von Mangoldt function. The selected square-root-scale target is

$$\exists C > 0 \ \ \forall x \ge 2, \qquad |E_\psi(x)| \le C\sqrt{x}\,(\log x)^2. \tag{1}$$

The Lean development currently verifies the finite von Mangoldt representation of E_ψ and the right-half-plane logarithmic-derivative identity

$$\sum_{n=1}^{\infty} \frac{\Lambda(n)}{n^s} = -\frac{\zeta'(s)}{\zeta(s)}, \qquad \mathrm{Re}(s) > 1. \tag{2}$$

It also records the equivalence between RH and Equation (1) as a proposition to reconstruct, not as an axiom or completed theorem.

16.3 Multiplicity as analytic vanishing order

For $s \neq 1$, zeta is analytic at s. Its zero multiplicity is represented by the analytic order

$$m(s) := \mathrm{ord}_s(\zeta) \in \mathbb{N} \cup \{\infty\}.$$

The formal development proves:

$$\begin{aligned} \mathrm{Nontriv}(\rho) &\Longrightarrow m(\rho) \neq 0, \\ s \neq 1 &\Longrightarrow (m(s) = 0 \Longleftrightarrow \zeta(s) \neq 0), \\ s \neq 1 &\Longrightarrow m(s) \neq \infty. \end{aligned}$$

The last statement is important: it justifies converting the extended natural analytic order into an ordinary finite integer multiplicity. Its Lean proof propagates finite order from $s = 2$, where zeta is nonzero, across the connected punctured plane $\mathbb{C} \setminus \{1\}$.

16.4 Compact, multiplicity-aware zeta divisor

Let $K \subset \mathbb{C}$ avoid the pole:

$$K \subseteq \mathbb{C} \setminus \{1\}.$$

The local zeta divisor is the integer-valued locally finite function

$$D_K(s) := \mathrm{div}_K(\zeta)(s).$$

For $s \in K$, the divisor value agrees with the finite analytic vanishing order. Outside K it is zero. The following properties are kernel-checked:

1. if K is compact, then $\mathrm{supp}(D_K)$ is finite;
2. $D_K(s) \ge 0$ for every s, because zeta is analytic on K;
3. for $s \in K$,
$$s \in \mathrm{supp}(D_K) \quad \Longleftrightarrow \quad \zeta(s) = 0;$$
4. every selected nontrivial zero has strictly positive integer multiplicity; and
5. for any weight $w : \mathbb{C} \to \mathbb{C}$, the multiplicity-aware sum
$$\sum_{\rho \in \mathrm{supp}(D_K)} D_K(\rho) w(\rho)$$
is a genuine finite sum.

Pinned Mathlib exposes zeta as analytic on $\mathbb{C} \setminus \{1\}$ but does not currently expose, through the imports used by this programme, a global meromorphic-continuation theorem suitable for silently crossing the pole. The explicit pole-avoidance hypothesis is therefore a formal safety condition, not merely a presentation choice.

Representative Lean declarations are:

```
noncomputable def zetaDivisorOn (K : Set C) :
    Function.locallyFinsuppWithin K Z :=
  divisor riemannZeta K

theorem zetaDivisorOn_support_finite {K : Set C}
    (hK : IsCompact K) :
    (zetaDivisorOn K).support.Finite

theorem mem_compactZetaDivisorSupport_iff_zero
    {K : Set C} (hK : IsCompact K)
    (havoid : K subset {1}^c) {s : C} (hs : s in K) :
    s in compactZetaDivisorSupport K hK <-> riemannZeta s = 0
```

As elsewhere in this report, ASCII placeholders in listings stand for the corresponding Unicode Lean symbols used in the checked source.

16.5 Pole-free compact exhaustion

For $n \in \mathbb{N}$, define

$$K_n := \overline{B}(0, n) \setminus B\left(1, \frac{1}{n+1}\right). \tag{3}$$

Each K_n is compact and excludes 1. The shrinking deleted ball is necessary because a plain expanding closed ball eventually contains the pole, where the available analytic divisor theorem cannot be applied.

The formalization proves the exhaustion property

$$s \neq 1 \quad \Longrightarrow \quad \exists n \in \mathbb{N},\ s \in K_n. \tag{4}$$

Filtering the finite divisor support by the nontrivial-zero predicate gives

$$\mathcal{Z}_n := \{\rho \in \operatorname{supp}(D_{K_n}) : \operatorname{Nontriv}(\rho)\}.$$

The sets $\mathcal{Z}_n$ are finite, and every nontrivial zero belongs to at least one of them. Thus the formal truncation cannot permanently omit an arbitrary fixed nontrivial zero.

16.6 Finite zero contribution and the exact open target

The zero kernel required by the classical explicit formula is

$$W_x(\rho) := \frac{x^\rho}{\rho}.$$

Using the compact divisor, the checked finite contribution is

$$Z_n(x) := \sum_{\rho \in \mathcal{Z}_n} D_{K_n}(\rho) \frac{x^\rho}{\rho}. \tag{5}$$

This construction is multiplicity-aware by definition; repeated zeros do not need to be represented by duplicated list entries.

For the current convention and nonintegral $x > 1$, define the elementary pole/trivial-zero correction

$$A(x) := \log(2\pi) + \frac{1}{2}\log\left(1 - x^{-2}\right). \tag{6}$$

The next analytic theorem has now been recorded exactly as `ChebyshevExplicitFormulaTarget`:

$$\forall x > 1, \quad x \notin \mathbb{N} \quad \Longrightarrow \quad Z_n(x) \longrightarrow -E_\psi(x) - A(x). \tag{7}$$

Equivalently, once convergence and conventions are justified,

$$E_\psi(x) = -\lim_{n\to\infty} Z_n(x) - A(x).$$

The nonintegrality hypothesis avoids selecting a left-, right-, or half-weight value at jump points of ψ. A later extension may introduce the standard symmetrized convention at prime-power jumps.

Open gate

Equation (7) is a typed Lean proposition, not a proved theorem. Defining $Z_n(x)$ and proving that its summands are genuine zeros does not establish convergence, justify a contour shift, or calculate the required residues. Those are the current mathematical proof obligations.

16.7 Completed specialized Perron interface

The first Perron unit has now been constructed in `RiemannLab/Tasks/PerronFormula.lean`. Define the zeta logarithmic-derivative kernel

$$\mathcal{K}_\zeta(x,s) := -\frac{\zeta'(s)}{\zeta(s)}\frac{x^s}{s} \tag{8}$$

and the equivalent von Mangoldt L-series kernel

$$\mathcal{K}_\Lambda(x,s) := \left(\sum_{n=1}^{\infty}\frac{\Lambda(n)}{n^s}\right)\frac{x^s}{s}. \tag{9}$$

For $\mathrm{Re}(s) > 1$, the previously verified logarithmic-derivative identity gives the kernel-checked equality

$$\mathcal{K}_\Lambda(x,s) = \mathcal{K}_\zeta(x,s). \tag{10}$$

For $c > 1$ and finite height T, parametrize the vertical segment by $s = c + it$. Since $ds = i\,dt$, the factor i cancels against the i in $(2\pi i)^{-1}$. The formal normalized truncations are therefore

$$P_T^\zeta(c,x) := \frac{1}{2\pi}\int_{-T}^{T}\mathcal{K}_\zeta(x,c+it)\,dt, \tag{11}$$

$$P_T^\Lambda(c,x) := \frac{1}{2\pi}\int_{-T}^{T}\mathcal{K}_\Lambda(x,c+it)\,dt. \tag{12}$$

Lean proves that every point $c + it$ remains in the half-plane of absolute convergence and hence, for every finite T,

$$c > 1 \quad\Longrightarrow\quad P_T^\Lambda(c,x) = P_T^\zeta(c,x). \tag{13}$$

The zero-height specializations are also checked:

$$P_0^\zeta(c,x) = P_0^\Lambda(c,x) = 0.$$

Representative Lean declarations are:

```
noncomputable def zetaPerronKernel (x : R) (s : C) : C :=
  (-deriv riemannZeta s / riemannZeta s) * (x : C) ^ s / s

noncomputable def truncatedZetaPerronIntegral
    (c T x : R) : C :=
  (1 / (2 * (Real.pi : C))) *
    integral t in -T..T, zetaPerronKernel x (c + t * Complex.I)

theorem truncatedVonMangoldtPerronIntegral_eq_zeta
    {c T x : R} (hc : 1 < c) :
    truncatedVonMangoldtPerronIntegral c T x =
      truncatedZetaPerronIntegral c T x
```

The listing uses portable ASCII placeholders; the checked source uses Lean's native Unicode notation and interval integral syntax.

16.7.1 Exact inversion target

For nonintegral $x > 1$, the classical Perron conclusion required by the programme is

$$\forall c > 1, \qquad \lim_{T\to\infty} P_T^\zeta(c,x) = \psi(x). \tag{14}$$

This is represented by the Lean proposition `ChebyshevPerronInversionTarget`, and is now proved by `chebyshevPerronInversion`. The proof passes through the arithmetic and zeta forms using Equation (13).

Open gate

Perron inversion is complete away from jumps. The theorem deliberately assumes x is not a natural number, so no half-weight convention at a jump of ψ is being smuggled into the result.

16.7.2 Pinned-library capability audit

The dependency audit found the following usable primitives in pinned Mathlib:

Module (prefix `Mathlib.`)	Available role
`Analysis.MellinTransform`	Mellin transform, inverse Mellin transform, and vertical integrability.
`Analysis.MellinInversion`	General Mellin inversion through Fourier inversion under its stated integrability and continuity hypotheses.
`Analysis.Complex.CauchyIntegral`	Rectangle boundary integrals and Cauchy–Goursat machinery needed for later contour shifting.
`NumberTheory.LSeries.Dirichlet`	Absolute convergence of the von Mangoldt L-series for $\mathrm{Re}(s) > 1$ and its identity with $-\zeta'/\zeta$.

No specialized arithmetic Perron theorem matching Equation (14) was found. This negative result is recorded in `perron_dependency_audit_v1.json`; it prevents the programme from treating a general Mellin inversion theorem as though all hypotheses for the discontinuous arithmetic step function had already been discharged.

16.8 Remaining analytic bridge

The Perron interface and the finite-height arithmetic/zeta identification are now complete. The shortest credible path from this interface to Equation (7) is decomposed as follows.

1. **Scalar Perron kernel limit.** For $c > 0$, $y > 0$, and $y \neq 1$, prove

$$Q_T(c,y) := \frac{1}{2\pi}\int_{-T}^{T} \frac{y^{c+it}}{c+it}\,dt \longrightarrow \begin{cases} 1, & y > 1, \\ 0, & 0 < y < 1. \end{cases} \tag{15}$$

   The omitted point $y = 1$ is the half-weight discontinuity. This isolates the Fourier/Mellin convergence issue from all arithmetic-series bookkeeping.

2. **Finite Dirichlet-polynomial inversion.** Apply Equation (15) term by term to a finite sum

$$\sum_{n \le N} \frac{\Lambda(n)}{n^s},$$

   producing the corresponding finite partial sum of $\psi(x)$.

3. **Absolute tail control.** Use absolute convergence on $\mathrm{Re}(s) = c > 1$ to control the omitted Dirichlet-series tail uniformly enough to pass from finite polynomials to Equation (14).

4. **Rectangular contour shift.** Move the vertical line from $\mathrm{Re}(s) = c$ into the left half-plane while avoiding zeros and the pole. Formal obligations include path composition, orientation, integrability, horizontal-edge bounds, and a sequence of admissible heights.

5. **Residue accounting.** Prove that the pole at $s = 1$ supplies the main term x, each nontrivial zero ρ supplies $-m(\rho)x^\rho/\rho$, and the remaining pole/kernel and trivial-zero contributions combine into Equation (6).

6. **Limit and ordering control.** Relate the zeros encountered by rectangular contours to the compact exhaustion K_n, prove the required convergence mode, and justify the symmetric ordering of the conditionally convergent zero contribution.

7. **RH-equivalent estimates.** Only after the explicit formula is established should the programme prove RH implies Equation (1), and conversely use the psi bound plus analytic continuation to exclude zeros with real part greater than 1/2.

Items 1–3 are complete. The irreducible unit is now the global punctured rectangle theorem: excise finitely many poles, relate the small-circle integrals to the checked local residues, and prove the displaced vertical and horizontal edges vanish along admissible heights.

Programme decision

The specialized Perron representation and its limit to $\psi(x)$ are checked. The next work must complete genuine pole excision and boundary estimates; a formal residue ledger alone is not a contour-shift theorem.

16.9 Sharp convergence and absolute Fourier reduction completed

The conditional-convergence obstacle has now been separated from the value calculation in `RiemannLab/Tasks/OscillatoryCauchy.lean`. For $c > 0$ and $a \neq 0$, define

$$H_T(c, a) := \frac{1}{2\pi} \int_{-T}^{T} \frac{e^{iat}}{c + it}\, dt.$$

Lean verifies an exact finite-interval integration-by-parts identity, proves that both endpoint terms vanish like $O((|a|T)^{-1})$, and proves absolute integrability of the squared-denominator remainder. Consequently the sharp symmetric truncations $H_T(c, a)$ genuinely converge; this is no longer an open assumption.

The same module now proves the source-closed Laplace identity

$$\frac{1}{c + it} = \int_0^\infty e^{-(c+it)u}\, du$$

and the exact reduction

$$\lim_{T\to\infty} H_T(c, a) = \frac{1}{2\pi a} \int_{-\infty}^{\infty} \frac{e^{iat}}{(c + it)^2}\, dt. \tag{16}$$

The integral on the right of Equation (16) is absolutely convergent. Thus the remaining analytic gate has the single named statement

$$\int_{-\infty}^{\infty} \frac{e^{iat}}{(c + it)^2}\, dt = \begin{cases} 2\pi a e^{-ac}, & a > 0, \\ 0, & a < 0. \end{cases} \tag{17}$$

Lean also verifies that Equation (17) implies the original oscillatory Cauchy target and hence the away-from-jump scalar Perron limit. The preferred next route is Fourier inversion for the

continuous one-sided function $u \mapsto ue^{-cu}\mathbf{1}_{(0,\infty)}(u)$, whose transform has precisely the squared denominator in Equation (17).

Verified result

At this intermediate checkpoint, sharp truncation, endpoint decay, and conditional tail control reduced the scalar Perron gate to one absolutely convergent Fourier value. The following section records the subsequent closure of that value.

16.10 Fourier value and scalar Perron theorem completed

The final scalar value has now been kernel-checked in `RiemannLab/Tasks/SquaredCauchyFourier.lean`. For $c > 0$, Lean constructs the continuous one-sided exponential moment

$$f_c(u) = \begin{cases} ue^{-cu}, & u > 0, \\ 0, & u \le 0, \end{cases}$$

proves its whole-line integrability, computes its Fourier transform, verifies the integrability of that transform, and applies the pinned Mathlib Fourier inversion theorem. This proves Equation (17) and propagates through the prior reductions to establish

$$\frac{1}{2\pi}\int_{-T}^{T} \frac{y^{c+it}}{c+it}\,dt \longrightarrow \begin{cases} 1, & y > 1, \\ 0, & 0 < y < 1, \end{cases}$$

for every $c > 0$ away from the jump $y = 1$. The scalar Perron target is therefore no longer open.

16.11 Finite von Mangoldt and Chebyshev inversion completed

The new module `RiemannLab/Tasks/FinitePerron.lean` lifts the scalar result to an arbitrary finite coefficient family. In particular, for nonintegral $x > 0$ it proves the finite von Mangoldt specialization

$$\sum_{n \le \lfloor x \rfloor} \Lambda(n) \frac{1}{2\pi}\int_{-T}^{T} \frac{(x/n)^{c+it}}{c+it}\,dt \longrightarrow \psi(x). \tag{18}$$

All sums in Equation (18) are finite, so this theorem contains no hidden exchange of an infinite series and an integral.

The full Perron module also proves fixed-height interchange of the absolutely convergent von Mangoldt series and the vertical integral. A height-independent Dirichlet bound supplies a summable coefficient majorant, so Tannery's theorem passes the growing-height limit through the series. Thus the complete zeta Perron integral converges to $\psi(x)$ away from integer jumps.

Verified result

The scalar theorem, finite inversion, fixed-height interchange, uniform coefficient domination, and the complete Chebyshev Perron formula are all kernel-checked. This is significant formalization of classical analysis, not a new theorem about zeta zeros and not a proof of RH.

16.12 Formal artifact map

Artifact	Technical role
`RiemannLab/Tasks/NontrivialZeros.lean`	Trivial/nontrivial predicates, critical-strip classification, reflection, conjugation, four-point orbit, and RH equivalence.
`RiemannLab/Tasks/CriticalLineCriterion.lean`	Chebyshev remainder, square-root-scale criterion, finite von Mangoldt sum, and zeta logarithmic-derivative bridge.
`RiemannLab/Tasks/ZetaZeroMultiplicity.lean`	Analytic zero order, analyticity at nontrivial zeros, and zero/nonzero characterization away from the pole.
`RiemannLab/Tasks/CompactZetaDivisor.lean`	Locally finite integer divisor, compact support, finite order, support/zero equivalence, nonnegativity, and finite weighted sums.
`RiemannLab/Tasks/ExplicitFormula.lean`	Pole-free compact exhaustion, finite nontrivial-zero sets, positive multiplicities, finite x^ρ/ρ sums, and the open convergence target.
`RiemannLab/Tasks/PerronFormula.lean`	Normalized zeta and von Mangoldt Perron kernels, finite-height vertical integrals, exact equality on $\mathrm{Re}(s) > 1$, fixed-height interchange, summable uniform majorant, Tannery limit, and full Chebyshev Perron formula.
`RiemannLab/Tasks/ContourShift.lean`	Oriented rectangle boundary, normalized right-edge identity, local multiplicity-aware zero residues, and residues at $s = 1$ and $s = 0$.
`RiemannLab/Tasks/ScalarPerron.lean`	Scalar kernel, logarithmic reduction, half-weight value at the jump, and exact equivalence with the oscillatory Cauchy limit.
`RiemannLab/Tasks/OscillatoryCauchy.lean`	Sharp convergence, integration-by-parts tail control, Laplace representation, and the absolute squared-kernel reduction subsequently discharged by Fourier inversion.
`RiemannLab/Tasks/SquaredCauchyFourier.lean`	One-sided exponential moment, Fourier transform and inversion, exact squared-Cauchy value, and completed scalar Perron theorem.
`RiemannLab/Tasks/FinitePerron.lean`	Generic finite coefficient inversion and finite von Mangoldt specialization to the Chebyshev function away from integer jumps.
`explicit_formula_dependency_graph_v1.json`	Machine-readable dependency state and the precommitted next proof unit.
`perron_dependency_audit_v1.json`	Pinned Mathlib capability audit, missing specialized theorem, and scalar Perron-kernel decomposition.

Verified result

The compact divisor and complete Chebyshev Perron formula are no longer blockers. Local zero residues with full multiplicity, the residues at $s = 1$ and $s = 0$, rectangle orientation, and the right-edge normalization are also verified. The remaining classical blocker is global pole excision together with trivial-zero summation and vanishing-edge estimates. After the explicit formula, the programme reaches the exact classical RH frontier.

17 Full Perron inversion and initial contour residues completed

The module `RiemannLab/Tasks/PerronFormula.lean` now proves

$$\lim_{T\to\infty}\frac{1}{2\pi}\int_{-T}^{T}-\frac{\zeta'(c+it)}{\zeta(c+it)}\frac{x^{c+it}}{c+it}\,dt=\psi(x) \tag{19}$$

for $c>1$, $x>1$, and nonintegral x. The decisive new estimate bounds each normalized coefficient integral by a summable majorant independent of T; Tannery's theorem then passes the limit through the infinite von Mangoldt series.

The new module `RiemannLab/Tasks/ContourShift.lean` fixes the four-side orientation of the rectangle and proves that its normalized right edge is exactly the established finite-height Perron integral. Analytic factorization at a zero gives the multiplicity-aware limit

$$(s-\rho)\frac{\zeta'(s)}{\zeta(s)}\longrightarrow m(\rho), \tag{20}$$

and therefore the contour kernel has local residue $-m(\rho)x^{\rho}/\rho$. The compact divisor converts these local values into a finite residue ledger with the same multiplicities used by the truncated zero sum. Mathlib's Laurent asymptotic at $s=1$ yields the main term x, while the known values $\zeta(0)=-1/2$ and $\zeta'(0)=-\log(2\pi)/2$ yield the residue $-\log(2\pi)$ at the Perron pole.

These are classical results formalized with kernel checking. They do not yet constitute the explicit formula: admissible-height estimates for the remaining three contour edges remain open.

18 Global residue theorem, trivial-zero sum, and final assembly

The classical contour infrastructure has advanced through three further machine-checked gates. The module `RiemannLab/Tasks/TrivialZeroResidues.lean` proves that every negative-even zero $-2(n+1)$ is simple. It identifies its contour residue as

$$-\frac{x^{-2(n+1)}}{-2(n+1)}=\frac{x^{-2(n+1)}}{2(n+1)}$$

and evaluates the full series as

$$\sum_{n\ge 0}\frac{x^{-2(n+1)}}{2(n+1)}=-\frac{1}{2}\log(1-x^{-2}),\qquad x>1.$$

Thus the trivial-zero simplicity and summation gate is complete.

The module `RiemannLab/Tasks/RectangularResidues.lean` proves the missing rectangular Cauchy kernel identity directly in the exact four-edge orientation used by the project:

$$\int_{\partial R}\frac{ds}{s-p}=2\pi i\qquad (p\in\operatorname{int}R).$$

The proof controls the branch cut by using $\log(s-p)$ on three sides and $\log(-(s-p))$ on the left side. It then lifts this identity to a finite global residue theorem. For a holomorphic remainder g, a finite pole set P, pole locations p_j, and residues r_j, Lean proves

$$\int_{\partial R}\left(g(s)+\sum_{j\in P}\frac{r_j}{s-p_j}\right)ds=2\pi i\sum_{j\in P}r_j,$$

provided every p_j lies strictly inside R. Continuity supplies the edge integrability automatically. This closes the global finite residue-collection gate without assuming a residue theorem absent from the pinned Mathlib version.

Finally, `RiemannLab/Tasks/ExplicitFormulaAssembly.lean` defines the exact remaining admissible-contour limit proposition and proves that it implies `ChebyshevExplicitFormulaTarget`. In particular, once a sequence of admissible heights tends to infinity, the normalized top, bottom, and displaced left edges tend to zero, and the finite residue ledger is instantiated, no further complex analysis or limit algebra remains: the already proved Perron limit forces the complete Chebyshev explicit formula.

The full pinned build at that checkpoint passed in 3,593 jobs. Two tightly coupled obligations remain: instantiate the finite principal-parts theorem for the actual zeta kernel on each chosen rectangle, and prove the zeta logarithmic-derivative estimate needed to construct admissible heights and make the three displaced edge integrals vanish. The first is now bounded analytic packaging around the proved residue theorem; the second is the substantive remaining classical estimate. Pinned Mathlib supplies neither that growth estimate nor a ready theorem from which it can be obtained. This is still established classical analytic number theory being formalized; it is not new evidence for RH and imposes no new constraint on the real parts of nontrivial zeros.

19 Finite principal parts for the actual zeta kernel

The analytic-packaging obligation from the preceding section is now complete. The new module `RiemannLab/Tasks/MeromorphicPrincipalPart.lean` proves a generic removal theorem. If F is meromorphic at p and

$$(s-p)F(s) \longrightarrow r \qquad (s \to p,\ s \neq p),$$

then there is a function analytic at p which agrees punctured-locally with $F(s) - r/(s-p)$. The proof is carried out through Mathlib's meromorphic order: multiplication by $(s-p)$ raises the order by one, and convergence to zero forces the remaining order to be nonnegative.

The module `RiemannLab/Tasks/FinitePrincipalParts.lean` lifts this local statement to an arbitrary finite pole set P. Given residues r_p, it constructs a single function g analytic at every $s \in \mathbb{C}$ such that

$$F(s) = g(s) + \sum_{p \in P} \frac{r_p}{s-p}, \qquad s \notin P.$$

The construction patches the removable values at the finitely many poles and proves analyticity both at each patched point and on the complement. Thus the finite residue theorem can now be applied without assuming that independently chosen local extensions form a global holomorphic remainder.

The module `RiemannLab/Tasks/ZetaContourPrincipalParts.lean` instantiates this theorem for the actual contour kernel

$$K_x(s) = -\frac{\zeta'(s)}{\zeta(s)} \frac{x^s}{s}.$$

It proves global meromorphy of ζ from its completed-zeta representation at $s = 1$ and analyticity elsewhere, hence global meromorphy of K_x. The principal parts are then removed at all four pole classes:

point	$\operatorname{Res} K_x$
1	x
0	$-\log(2\pi)$
$-2(n+1)$	$-x^{-2(n+1)}/[-2(n+1)]$
ρ nontrivial	$-m(\rho)x^\rho/\rho.$

For any finite ledger whose entries carry one of these classifications, and which contains every singularity under discussion, Lean now produces the globally analytic remainder required by the rectangular residue theorem.

20 Quantitative reduction of the contour edges

The new module `RiemannLab/Tasks/ContourEdgeEstimates.lean` isolates the remaining large-height input by proving the exact identity

$$\|K_x(s)\| = \left\|\frac{\zeta'(s)}{\zeta(s)}\right\| \frac{x^{\Re s}}{\|s\|}, \qquad x > 0.$$

It then converts any supplied bound $\|\zeta'(s)/\zeta(s)\| \le L$ and lower bound $R \le \|s\|$ into

$$\|K_x(s)\| \le \frac{L x^{\Re s}}{R}.$$

Uniform pointwise bounds are lifted to the exact horizontal and vertical interval integrals used by the project. In particular, if C_n uniformly bounds a horizontal edge from a to b and $C_n|b-a| \to 0$, Lean proves that the corresponding complex edge integral tends to zero.

The full pinned build now passes in 3,599 jobs. The finite principal-parts instantiation is therefore closed. What remains is the substantive classical large-height theorem: construct heights avoiding zero ordinates with a uniform bound for ζ'/ζ, discharge the top and bottom edges through the new norm estimates, and combine the functional equation and gamma-factor bounds with a displaced-left sequence to make the left edge vanish. The finite ledger must finally be identified with the compact multiplicity-aware zero truncation already used by the target statement.

Pinned Mathlib does not contain this global admissible-height estimate. No complete Chebyshev explicit formula is claimed at this checkpoint. All new results remain formalizations of classical analytic infrastructure and provide no new restriction on nontrivial zero locations.

A Key evidence hashes

Artifact	SHA-256
Immutable paid live result	`5795e3faa55cf2a7466a3537ac41005056d44711eaa826c84935af318437b2e6`
Established zeta baseline	`d121313f104e8e9d3a62fd2270e46aa6c4a7c6888ebc7d996f004fae5a25943c`
Catalogue probe	`6691c0dff28a5b761b7c4d1da63f7820cb0b57976b21979594be093e149fd5b5`
V2 packet	`2190c93a840eebabf0ad7cbfc3dbfe53a01ca32ead4f56765143d190e50d3660`
Sealed v3 packet	`1771f8b9977ce0cf1ffac73b1284361e61ae92d4f86672afe97a241a786abb6e`
Completed v3 result	`52899a504530456f1c6f90acf28a34edddc54ccbb4ce765d145c2f9166bc951d`
Pinned v2 environment	`6776200251cfd041ecda62038ae8ff02980bf1066a332289b6c96ca015485591`

The local-verification report is regenerated and therefore receives a new hash when timestamps or verified memory state are rebuilt. The timestamped paid result is the immutable historical object.

B V2 statement hashes

Ref.	Statement SHA-256
T1	`f84686a87dff3c814d79a296232fad81531194b318c4c81fe7d5aab8c9050798`
T2	`d1ca22f154945c4c26cf5f5cecfc5bf86d52d1538375d94420ea5abf7fce2760`
T3	`3c6a7aa1eede91cbd24530d477c38f750176269d1d8c08773dd2bc510fe77d31`
T4	`37c2b2c0f357b924d945d3c815244f0eff7affc33372d60ba839caea750cace5`
T5	`fed50dcca60493a2beedc23cb7ea0d485c1659dbe0fabdd0b8d3e698db523d3a`
T6	`6e920be25538210c2ec4850febc0c8a50e426af5c4ba743253c84b8af3611b4b`

C Reproducibility commands

Run from the COMDEX repository root:

```
.venv/bin/python scripts/aion_riemann_local_verify.py
.venv/bin/python -m pytest backend/tests/test_aion_riemann_lab.py -q
```

The v3 live comparison and the delayed retention comparison are complete and protected by their one-shot execution receipts. Neither result established an AION advantage, and no further paid run is scheduled by this document.

D Selected references

References

[1] B. Riemann, “On the Number of Prime Numbers Less Than a Given Quantity,” 1859.

[2] E. C. Titchmarsh, revised by D. R. Heath-Brown, *The Theory of the Riemann Zeta-Function*, Oxford University Press.

[3] H. M. Edwards, *Riemann’s Zeta Function*, Dover Publications.

[4] Lean Prover community, *Lean 4 theorem prover and reference documentation.*

[5] Mathlib community, *Mathlib: a mathematical library for Lean*, pinned in this programme at v4.32.1.

[6] AION/COMDEX, *Riemann Hypothesis Programme continuation handover, local verification report, task packets, Lean sources, and immutable JSON evidence*, 14–15 August 2026.